

Foundations of Audio Signal Processing

§6 Systems and Filters

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WINTER TERM

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Introducing Remarks

- Up to now, we mainly analyzed signals by calculating their representations with respect to suitable Hilbert bases:
 - ◆ [Fourier series](#), [Fourier-Transform](#) (later chapter: [Wavelet-Transform](#)).
- This kind of analysis allows us to obtain important information about the original signal, e.g., about the contained harmonic oscillations. All of the resulting new representations are lossless: The original signal may be perfectly reconstructed.
- It is not changed but only transferred to an alternative representation.
- However, many applications [require to change](#) the original signal:
 - ◆ [Attenuation or emphasis of spectral characteristics](#), e.g. for [noise reduction](#),
 - ◆ [Compression, source separation, time stretching](#), ...
- In this chapter we introduce fundamental mechanisms for manipulating signals.

Linear systems

- In general, every mechanism for modifying a signal is called a system.
- Formally, a **system** is specified by a mapping $T : E \rightarrow A$, transforming an input signal $x \in E$ to an output signal $y \in A$. In this, E and A denote suitable signal spaces.

$$x \longrightarrow \boxed{\text{System } T} \longrightarrow y.$$

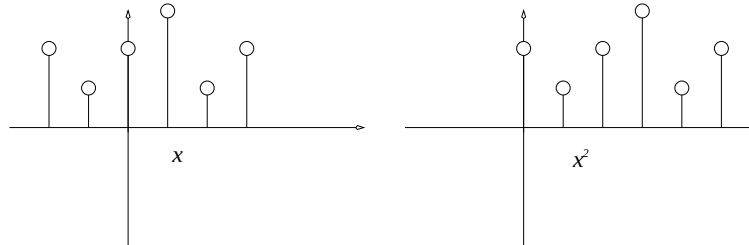
- Here, we will only consider DT-signals. The signal spaces will be the previously introduced $\ell^p(\mathbb{Z})$, $p \in [1, \infty]$.
- The simplest, yet very important, class of systems are the **linear systems**. Here, the spaces E and A are linear while the mapping $T : E \rightarrow A$ is $\mathbb{C}$ -linear, i.e. for all $x, y \in E$ and all $\lambda \in \mathbb{C}$:

$$T[x + y] = T[x] + T[y] \quad \text{and} \quad T[\lambda x] = \lambda T[x].$$

- We use the mathematical term **operator** interchangeably with the term system.

Time-shifts are linear operators

- The **time-shift operator** T^k , $k \in \mathbb{Z}$, is defined by $T^k[x](n) := x(n - k)$.
- It is easy to see that for every $p \in [1, \infty]$, T^k is a linear operator from $\ell^p(\mathbb{Z})$ to $\ell^p(\mathbb{Z})$ which conserves the norm.
- We will frequently write x^k instead of $T^k[x]$. Particularly, $x^0 = x$ and δ^k is the indicator function for $k \in \mathbb{Z}$, i.e. $\delta^k(j) = \delta_{k,j}$ for $j \in \mathbb{Z}$.
- The following figure illustrates the operator T^2 :

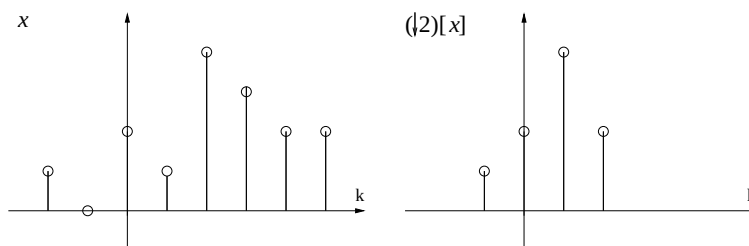


Downsampling-operators are linear

- For $M \in \mathbb{Z}_{\geq 2}$, we define the **(M-) downsampling-operator** $\downarrow M$ by

$$(\downarrow M)[x](n) := x(M \cdot n).$$

- It is easy to see that for every $p \in [1, \infty]$, $\downarrow M$ is a linear operator from $\ell^p(\mathbb{Z})$ to $\ell^p(\mathbb{Z})$.
- However, this operator is generally **lossy**.
- Downsampling only conserves each M -th value of an input signal:

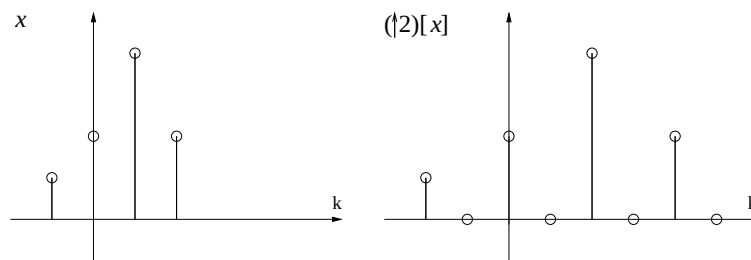


Upsampling-operators are linear

- For $M \in \mathbb{N}$, the **(M-) upsampling-operator** $\uparrow M$ is defined by

$$(\uparrow M)[x](n) = \begin{cases} x(n/M), & \text{if } M|n, \\ 0, & \text{otherwise.} \end{cases}$$

- It is easy to see that for every $p \in [1, \infty]$, $\uparrow M$ is a linear operator conserving both information and norm of the input signal.
- This operator temporally stretches the input signal x by a factor of M and inserts zeros at all resulting $M - 1$ positions between two original x -values:



Amplitude modulation is a linear operator

- **Amplitude modulation** with respect to a carrier signal $c \in \ell^\infty(\mathbb{Z})$ is a linear operator defined by $m_c[x](n) := c(n) \cdot x(n)$.
- An important special case is the **frequency-shift** E_ω , $\omega \in (0, 1)$, defined by

$$E_\omega[x](n) := e^{-2\pi i \omega n} x(n).$$

- For the case $x = f_{\omega_0} = (e^{2\pi i \omega_0 n})_{n \in \mathbb{Z}} \in \ell^\infty(\mathbb{Z})$ one obtains $E_\omega[f_{\omega_0}] = f_{\omega_0 - \omega}$.
- The operators E_ω and $E_{-\omega}$ are inverse to each other with respect to operator composition: $(E_{-\omega} \circ E_\omega)[x] = x$.
- The operator $E_{-\omega}$ is therefore called **demodulation operator**.
- Application: **frequency multiplexing**, e.g., for communication signals.
- ... or to make a bat sound audible to humans!

Rounding operators are not linear

- The **rounding operators** defined by the

$$\begin{aligned}\text{Ceil-operator:} \quad & \lceil x \rceil(n) := \lceil x(n) \rceil, \quad \text{für } x: \mathbb{Z} \rightarrow \mathbb{R} \\ \text{Floor-operator:} \quad & \lfloor x \rfloor(n) := \lfloor x(n) \rfloor, \quad \text{für } x: \mathbb{Z} \rightarrow \mathbb{R}\end{aligned}$$

are not linear.

- However, those operators are **time-invariant**:

$$\lceil x^k \rceil = \lceil x \rceil^k, \quad \lfloor x^k \rfloor = \lfloor x \rfloor^k.$$

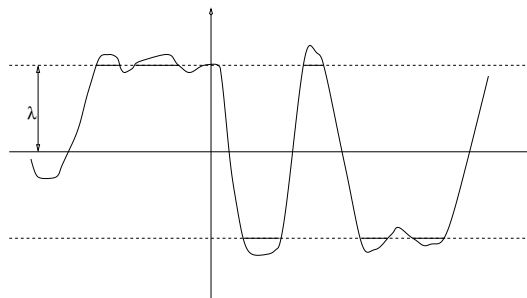
- Operators which are not linear are commonly called **non-linear**.

Cut-off-Operators are non-linear

- For $x: \mathbb{Z} \rightarrow \mathbb{C}$, the **cut-off operator** for $\lambda > 0$ is defined by

$$\text{Cut}_\lambda[x](n) := \begin{cases} x(n) & \text{if } |x(n)| \leq \lambda, \\ \frac{x(n)}{|x(n)|} \lambda & \text{otherwise.} \end{cases}$$

- Cut_λ is non-linear but time-invariant.
- In the following, a cut-off-operation on a CT-signal $x: \mathbb{R} \rightarrow \mathbb{R}$ is illustrated:



Continuous linear systems

- We are interested in linear systems with non-pathological behaviour. An important property of a system is continuity.
- A linear system $T: \ell^p(\mathbb{Z}) \rightarrow \ell^r(\mathbb{Z})$ is called **continuous**, if T transforms a convergent sequence of input signals into a convergent sequence of output signals.

Theorem. A continuous linear system $T: \ell^p(\mathbb{Z}) \rightarrow \ell^r(\mathbb{Z})$ may be interchanged with an infinite summation. Particularly, for $x \in \ell^p(\mathbb{Z})$ the following holds:

$$T\left[\sum_{k \in \mathbb{Z}} x(k)\delta^k\right] = \sum_{k \in \mathbb{Z}} x(k)T[\delta^k].$$

Proof. For $n \in \mathbb{N}$ let $\sigma_n := \sum_{|k| \leq n} x(k)\delta^k$. Then σ_n converges towards x with respect to the ℓ^p -norm. As T is continuous, $T[\sigma_n]$ then converges towards $T[x]$ with respect to the ℓ^r -norm. This is,

$\left\|T[x] - \sum_{|k| \leq n} x(k)T[\delta^k]\right\|_r \rightarrow 0$ for $n \rightarrow \infty$. Hence we obtain

$$T[x] = \lim_{n \rightarrow \infty} \sum_{|k| \leq n} x(k)T[\delta^k] = \sum_{k \in \mathbb{Z}} x(k)T[\delta^k]. \quad \square$$

LTI-systems

- Of particular importance are systems that commute with every time-shift T^k .
- A system $S: \ell^p(\mathbb{Z}) \rightarrow \ell^r(\mathbb{Z})$ is called **time invariant**, if $S \circ T^k = T^k \circ S$ holds for all $k \in \mathbb{Z}$. This means, for all $k \in \mathbb{Z}$ and all $x \in \ell^p(\mathbb{Z})$:

$$S[x^k] = S[x]^k.$$

- A linear, time-invariant system is also abbreviated as **LTI-system**.
- LTI-systems have an important characterization in terms of the convolution operator. This will be investigated now.

Convolution

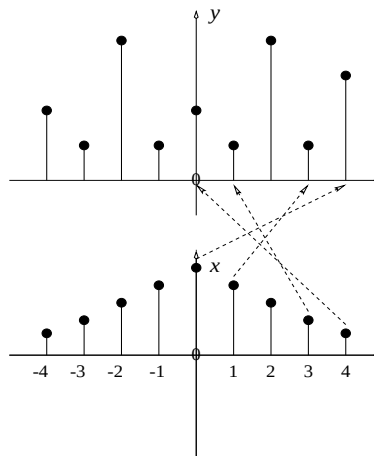
- The convolution is a kind of multiplication of two input signals that results in a single output signal. The convolution plays an essential role in describing filter operations. It is hence a fundamental tool in digital signal processing.
- **Remark:** It is important to **distinguish** the convolution from the **pointwise multiplication** of two signals. This multiplication is defined by $(x \cdot y)(n) := x(n) \cdot y(n)$.
- For two signals $x, y: \mathbb{Z} \rightarrow \mathbb{C}$, the **convolution** of x and y at position $n \in \mathbb{Z}$ is defined by

$$(x * y)(n) := \sum_{k \in \mathbb{Z}} x(k)y(n - k).$$

- **Note:** $x * y$ only exists if x and y fulfill suitable conditions, e.g. $x \in \ell^1(\mathbb{Z}), y \in \ell^\infty(\mathbb{Z})$.
- Further sufficient conditions guaranteeing the existence of $x * y$ are given in the next theorem. But first, the convolution is illustrated by an example.

Illustration of the convolution

Let $y: \mathbb{Z} \rightarrow \mathbb{C}$ denote a signal and let x be a probability density function on $\mathbb{Z}$, i.e. $x: \mathbb{Z} \rightarrow [0, 1]$ and $\sum_{n \in \mathbb{Z}} x(n) = 1$. Then $(x * y)(n) = \sum_{k \in \mathbb{Z}} x(k)y(n - k)$ may be interpreted as weighted average of y in a neighborhood of position n . The following figure gives an illustration for $n = 4$.



Properties of the convolution

Theorem. Let x, y, z denote signals $\mathbb{Z} \rightarrow \mathbb{C}$. Under the assumption that the following convolutions and products exist, the following laws hold:

- **Commutativity:** $x \cdot y = y \cdot x, \quad x * y = y * x,$
- **Associativity:** $(x \cdot y) \cdot z = x \cdot (y \cdot z), \quad (x * y) * z = x * (y * z),$
- **Distributivity:** $(x + y) \cdot z = x \cdot z + y \cdot z, \quad (x + y) * z = x * z + y * z.$

From this and the above theorem it follows that for conjugate exponents p and q and for a fixed $y \in \ell^q(\mathbb{Z})$ the definition

$$C_y[x] := x * y$$

yields a well-defined and linear **convolution operator** $C_y : \ell^p(\mathbb{Z}) \rightarrow \ell^\infty(\mathbb{Z})$. This operator, moreover, is Lipschitz-continuous. This stronger form of continuity requires that there exists a constant $c > 0$, such that $\|C_y[x] - C_y[x']\|_\infty \leq c \cdot \|x - x'\|_p$ holds.

(Exercise: Recall the definition of (classical) continuity. Show that Lipschitz-continuity implies continuity.)

Convolution theorem

Convolution operators have the following important property:

Convolution theorem. For $x, y \in \ell^2(\mathbb{Z})$ with $x * y \in \ell^2(\mathbb{Z})$,

$$\widehat{x * y} = \hat{x} \cdot \hat{y}$$

holds.

The Fourier transform of the convolution of two signals hence coincides with the pointwise multiplication of the Fourier transforms of both signals.

Remarks:

- The convolution theorem, again, only holds in the $L^2([0, 1])$ -sense. Particularly, in the general case $\widehat{x * y}(\omega) = \hat{x}(\omega) \cdot \hat{y}(\omega)$ only holds for almost all $\omega \in [0, 1]$.
- The convolution theorem is the basis for efficiently computing convolutions using FFTs.

Time-shifts are convolution operators

Theorem. For $k \in \mathbb{Z}$ the time-shift operator $T^k : \ell^\infty(\mathbb{Z}) \rightarrow \ell^\infty(\mathbb{Z})$ coincides with the convolution operator associated with the k -shifted delta-function δ^k :

$$T^k[x] = x^k = x * \delta^k, \quad \text{for all } x \in \ell^\infty(\mathbb{Z}).$$

Proof. For $n \in \mathbb{Z}$, the left hand side is $T^k[x](n) = x(n - k)$. For the corresponding value of the right hand side we obtain

$$\begin{aligned} (x * \delta^k)(n) &= \sum_{\ell \in \mathbb{Z}} x(\ell) \delta^k(n - \ell) \\ &= \sum_{\ell \in \mathbb{Z}} x(\ell) \delta_{k, n - \ell} = \sum_{\ell \in \mathbb{Z}} x(\ell) \delta_{\ell, n - k} \\ &= x(n - k). \end{aligned}$$

Hence we have proved that the time-shift operator T^k and the convolution operator C_{δ^k} coincide for $\ell^\infty(\mathbb{Z})$. $\square$

Continuous LTI-systems

The last theorem may be generalized as follows:

Theorem. Let $T : \ell^p(\mathbb{Z}) \rightarrow \ell^q(\mathbb{Z})$ denote a continuous LTI-system ($p, q \in [1, \infty]$). Then, T is the convolution operator of the so-called **impulse response** $h := T[\delta]$. This is, $T = C_h$ and for all $x \in \ell^p(\mathbb{Z})$, $T[x] = h * x$ holds.

Proof. As the unit impulse δ is contained in $\ell^p(\mathbb{Z})$, the sequence $h := T[\delta] \in \ell^q(\mathbb{Z})$ is well-defined. Furthermore, for all $n \in \mathbb{Z}$ and $x \in \ell^p(\mathbb{Z})$, it holds that

$$\begin{aligned} T[x](n) &= T\left[\sum_k x(k) \delta^k\right](n) \\ &= \sum_k x(k) T[\delta^k](n) \quad (\text{linearity and continuity of } T) \\ &= \sum_k x(k) T[\delta]^k(n) \quad (\text{time-invariance of } T) \\ &= \sum_k x(k) h(n - k) = (h * x)(n). \quad \square \end{aligned}$$

BIBO-stability

- By the last theorem, continuous LTI-systems correspond to convolution operators with the impulse response of the system.
- By imposing additional requirements on the LTI-system, one obtains an equivalent characterization of LTI-systems by convolution operators.
- A linear system $T: \ell^p(\mathbb{Z}) \rightarrow \ell^p(\mathbb{Z})$ is called **stable**, if T is continuous and for all $k \in \mathbb{Z}$, $T[\delta^k]$ is contained in $\ell^1(\mathbb{Z})$.
- In case that $p = \infty$, the system is called **BIBO-stable** (**B**ounded **I**ntput $\rightarrow$ **B**ounded **O**utput).

Characterization of stable LTI-systems

Theorem. For a linear system $T: \ell^p(\mathbb{Z}) \rightarrow \ell^p(\mathbb{Z})$, the following are equivalent:

- (1) T is a stable LTI-system.
- (2) There exists an $h \in \ell^1(\mathbb{Z})$ such that $T = C_h$, i.e., $T[x] = h * x$, for all $x \in \ell^p(\mathbb{Z})$.

Proof. (1) $\Rightarrow$ (2): This was already shown above.

(2) $\Rightarrow$ (1): The linearity is obvious. The time-invariance follows from the associativity of the convolution operator:

$$T[x^k] = h * (x^k) = h * (x * \delta^k) = (h * x) * \delta^k = T[x]^k.$$

The continuity follows from Young's inequality: As $x_m \rightarrow x$ in $\ell^p(\mathbb{Z})$ for $m \rightarrow \infty$ we obtain

$$\|h * (x - x_m)\|_p \leq \|h\|_1 \cdot \|x - x_m\|_p \rightarrow 0. \quad \square$$

FIR-, IIR- and causal filters

Let T denote a continuous LTI-system and $h := T[\delta]$.

- The sequence h is called **impulse response** and $h(n)$ is called **n -th filter coefficient** (or, sometimes **n -th filter tap**).
- T is called **FIR-filter** or **FIR-system** (Finite Impulse Response), if only finitely many filter coefficients are non-zero. Otherwise, T is called **IIR-filter** or **IIR-system** (Infinite Impulse Response).
- Usually, T and its impulse response h are identified. Then h is called filter although, strictly speaking, the associated convolution operator C_h is referred to.
- T is called **causal**, if $h(n) = 0$ for all $n < 0$. Causality is important in view of real-time applications (as one cannot access the future values of a signal).

Order and length of causal FIR-filters

- The **length** $\ell(h)$ of an FIR-filter h is defined as

$$\ell(h) := 1 + \max\{n | h(n) \neq 0\} - \min\{n | h(n) \neq 0\}.$$

- The length of the signal resulting from the convolution of two non-zero FIR-sequences x and y is given by $\ell(x * y) = \ell(x) + \ell(y) - 1$.
- Causal FIR-filters $T = C_h$ of **order** N and **length** $N + 1$ are filters that can be written as

$$T[x](n) = \sum_{\ell=0}^N h(\ell)x(n - \ell)$$

with filter coefficients $h(0), \dots, h(N)$, $h(N) \neq 0$ and $h(0) \neq 0$. The n -th value of the output signal $T[x]$ then only depends on the *past* values $x(n - 1), \dots, x(n - N)$ as well as on the *present* value $x(n)$ of the input signal x . (Hence one speaks of *causality*.) Those values of x are weighted by the filter coefficients and then added up to yield the value of the output signal.

Stable LTI-systems: (counter-) examples

- The **time-shift operators** T^k are stable LTI-systems, because $T^k = C_{\delta^k}$ and $\delta^k \in \ell^1(\mathbb{Z})$. Furthermore, all convolution operators are time-invariant and Lipschitz-continuous.
- The **downsampling operators** $(\downarrow M)$ are linear and continuous. Furthermore $(\downarrow M)$ is stable as for all k the sequence $y := (\downarrow M)[\delta^k]$ is contained in $\ell^1(\mathbb{Z})$. (If k is not a multiple of M , then y is identical to zero, otherwise $y = \delta^{k/M}$.) However, $(\downarrow M)$ is not time-invariant, as

$$((\downarrow M) \circ T^k)[x](n) = x(nM - k) \quad \text{but} \quad (T^k \circ (\downarrow M))[x](n) = x(M(n - k)).$$

- Analogously the **upsampling operators** $(\uparrow M)$ are linear, continuous and stable, but not time-invariant.
- The **frequency-shift operators** E_ω , $\omega \in [0, 1]$, defined by $E_\omega[x](n) := e^{-2\pi i \omega n} \cdot x(n)$ are linear and continuous. However, only for the case $\omega = 0$ the operator E_ω is time-invariant. (Exercise!)

The frequency response

- The spectral characteristics of LTI-systems are described by the Fourier transform of its impulse response.
- For a BIBO-stable LTI-system T with impulse response $h = T[\delta] \in \ell^1(\mathbb{Z})$, the Fourier transform

$$\hat{h}(\omega) := \sum_{n \in \mathbb{Z}} h(n) e^{-2\pi i n \omega}, \quad \omega \in [0, 1],$$

is called **frequency response** of T .

- **Notation:** Instead of $\hat{h}$ one usually writes H . Generally, lower case letters $f, g, h, \dots, x, y \dots$ are used to denote discrete filters and DT-signals. The corresponding Fourier transformed signals are then denoted by the corresponding capital letters $F, G, H, \dots, X, Y \dots$

Frequency responses as eigenvalues

Theorem. Let $h \in \ell^1(\mathbb{Z})$. Furthermore, let $f_\omega := (e^{2\pi i \omega k})_{k \in \mathbb{Z}} \in \ell^\infty(\mathbb{Z})$ denote the **frequency sequence** for $\omega \in [0, 1]$. Then, for each frequency ω the frequency sequence f_ω is an eigenvector of the convolution operator C_h for the eigenvalue

$$H(\omega) = \sum_{k \in \mathbb{Z}} h_k e^{-2\pi i k \omega} = \hat{h}(\omega), \quad \text{i.e.} \quad C_h[f_\omega] = H(\omega) f_\omega.$$

Therefore, $H(\omega)$ is called the **frequency response** of C_h for the frequency ω .

Proof. The claim follows from

$$\begin{aligned} C_h[f_\omega](n) &= \sum_{k \in \mathbb{Z}} h(k) f_\omega(n - k) = \sum_{k \in \mathbb{Z}} h(k) e^{2\pi i \omega (n - k)} \\ &= \sum_{k \in \mathbb{Z}} h(k) e^{-2\pi i \omega k} \cdot e^{2\pi i \omega n} = H(\omega) \cdot f_\omega(n). \quad \square \end{aligned}$$

BIBO-stable LTI-systems hence simply multiply the (normalized) frequency sequence of frequency ω by a factor of $H(\omega)$. For $|H(\omega)| > 1$ ($|H(\omega)| < 1$) the frequency is amplified (attenuated).

Amplitude- and phase response

Representing the frequency response $\hat{h}(\omega) = H(\omega) = \sum_{n \in \mathbb{Z}} h(n) e^{-2\pi i \omega n}$ of the convolution operator corresponding to $h \in \ell^1(\mathbb{Z})$ in polar coordinates

$$H(\omega) = |H(\omega)| \cdot e^{2\pi i \Phi_h(\omega)}$$

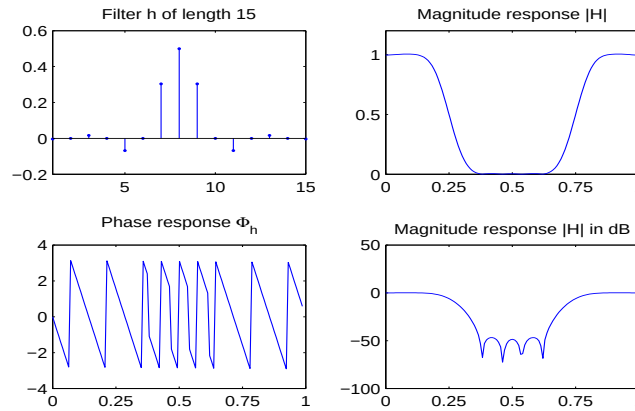
one obtains the

- **amplitude response** $\omega \mapsto |H(\omega)|$,
- **phase response** $\omega \mapsto \Phi_h(\omega)$ and
- the **spectral energy density** $\omega \mapsto |H(\omega)|^2$ of h .

Note that the phase response is only defined for frequencies ω where $H(\omega) \neq 0$.

Illustration: amplitude- & phase-response

The following illustration shows the amplitude response $|H|$ and the phase response Φ_h of a filter h of length 15. The filter coefficients $\neq 0$ of h are $h(1) = -0.0036 = h(15)$, $h(3) = 0.0161 = h(13)$, $h(5) = -0.0682 = h(11)$, $h(7) = 0.3038 = h(9)$ and $h(8) = 0.5$. The behaviour of $|H|$ is furthermore represented in the decibel-scale.



Properties of the frequency response

The frequency response H of a filter $h \in \ell^1(\mathbb{Z})$ contains the **spectral information** concerning h and has the following properties:

- For real-valued h , $\overline{H(\omega)} = H(-\omega)$. Hence the amplitude response then is an even function, $|H(\omega)| = |H(-\omega)|$, while the phase response is an odd function, $\Phi_h(-\omega) = -\Phi_h(\omega)$.
- The frequency response is 1-periodic. As the frequency response satisfies $H(\omega) = H(-\omega)$ in case of real-valued h , in this case all information on H is contained in the restriction of H to the interval $[0, \frac{1}{2}]$, as for $\omega \in [0, \frac{1}{2}]$ one obtains

$$H(1 - \omega) = H(-\omega) = \overline{H(\omega)}.$$

- Spectral information regarding low (high) frequencies correspond to the $\omega \in [0, \frac{1}{2}]$ near 0 (near $\frac{1}{2}$ respectively).

Low-pass, high-pass and band-pass filters (1)

According to the convolution theorem, the frequency content $X = \hat{x}$ of an input signal x after convolution by h is modified by a point-wise multiplication with the frequency response H . Depending on the form of the amplitude response $|H|$ one assigns filters to the categories of low-pass, high-pass and band-pass filters.

- $h \in \ell^1(\mathbb{Z})$ describes an (ideal) **low-pass filter** with a cut-off frequency of $\omega_0 \in [0, \frac{1}{2}]$, if low frequencies $\omega \in [0, \omega_0]$ pass the system without attenuation, while high frequencies $\omega \in [\omega_0, \frac{1}{2}]$ are cut-off. More precisely,

$$|H(\omega)| \approx \begin{cases} 1 & \text{if } |\omega| \in [0, \omega_0] \\ 0 & \text{if } |\omega| \in (\omega_0, \frac{1}{2}] \end{cases}$$

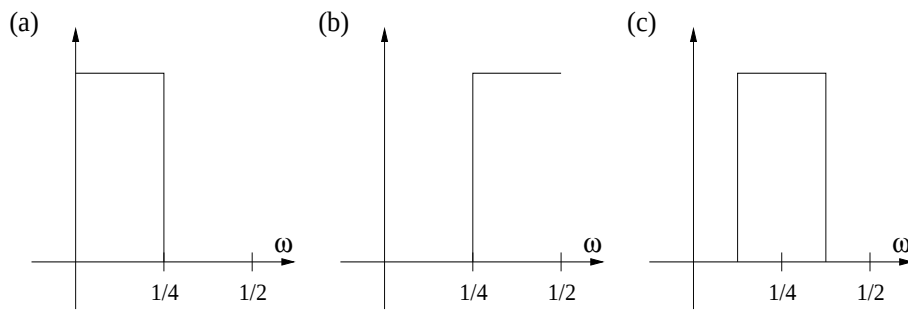
- Correspondingly, h is called an (ideal) **high-pass filter**, if

$$|H(\omega)| \approx \begin{cases} 1 & \text{for } |\omega| \in [\omega_0, \frac{1}{2}] \text{ and} \\ 0 & \text{for } |\omega| \in [0, \omega_0) \end{cases}$$

Low-pass, High-pass and band-pass filters (2)

- Finally, h is an (ideal) **band-pass filter** for the cut-off frequencies $0 < \omega_0 < \omega_1 < \frac{1}{2}$, if

$$|H(\omega)| \approx \begin{cases} 1 & \text{for } |\omega| \in [\omega_0, \omega_1] \text{ and} \\ 0 & \text{for } |\omega| \in [0, \omega_0) \cup (\omega_1, \frac{1}{2}] \end{cases}$$



Amplitude response of ideal (a) low-pass-, (b) high-pass-, and (c) band-pass filters

Low-pass, High-pass and band-pass filters (3)

Remarks:

- Note that besides those three categories there are several other filter types. For example, **notch-filters** are used to attenuate only certain (isolated) frequencies.
- Low- and high-pass filters may be regarded as special band-pass filters.
- Ideal band-pass filters cannot be realized by FIR systems and are not causal. Hence real systems have to rely on suitable approximations.

The z -transform

The values of discrete signals may be interpreted as coefficients of polynomials (finite signals) or formal power series (infinite signals) in some variable, say z .

- For example, $x = (1, 3, 5) \rightsquigarrow 1 \cdot z^1 + 3 \cdot z^2 + 5 \cdot z^3 =: \mathbf{x}(z)$.
- Then, the convolution rather obviously corresponds to a multiplication of polynomials.
- Furthermore, a substitution $z \mapsto e^{2\pi i \omega}$ naturally yields a connection to the Fourier spectrum.
- Such a representation is useful and frequently used in filter design.
- This is formalized by the notion of z -transform or Laurent-series.

Definition of the z -transform

Definition. Let $x : \mathbb{Z} \rightarrow \mathbb{C}$ denote a signal. The z -transform $Z[x] =: X$ of x is the function $X : D_x \rightarrow \mathbb{C}$ defined by

$$X(z) := \sum_{n \in \mathbb{Z}} x(n) z^{-n},$$

where D_x denotes the largest open subset of $\mathbb{C}$ such that for all $z \in D_x$ the sum on the right hand side exists.

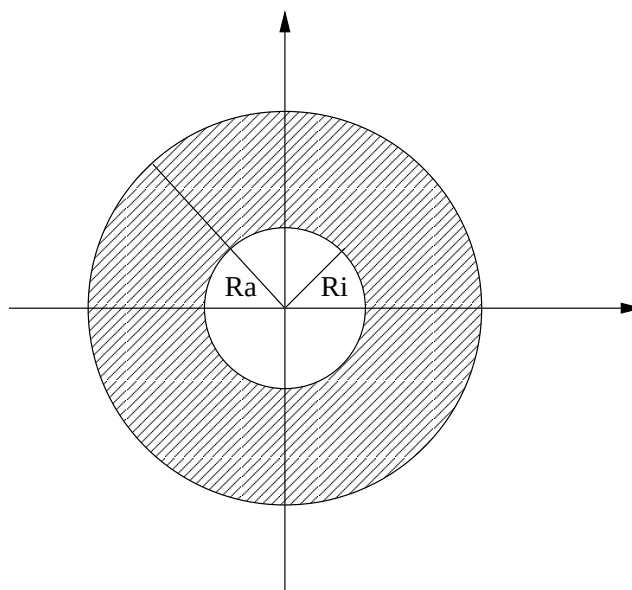
The region of convergence is disc-shaped. The inner radius is given by

$$R_i(x) = \overline{\lim}_{n \rightarrow \infty} |x(n)|^{1/n},$$

while the outer radius is

$$R_a(x) = \frac{1}{\overline{\lim}_{n \rightarrow \infty} |x(-n)|^{1/n}}.$$

Laurent series: region of convergence (illustration)



Properties of the z -transform

Theorem. Let x, y, h denote signals $\mathbb{Z} \rightarrow \mathbb{C}$ with associated z -transforms X, Y, H . Then the following properties hold:

1. *Linearity:*

$$Z(\lambda x + y) = \lambda Z(x) + Z(y)$$

(on the common region of convergence!)

2. *Delay-property:* Let $y = x^k = T^k[x]$ denote the k -shifted version of x , then

$$Y(z) = z^{-k} X(z).$$

3. *Multiplication by an exponential:* let $0 \neq a \in \mathbb{C}$ and $y(n) = a^n \cdot x(n)$, then $D_y = |a| \cdot D_x$ and

$$Y(z) = X(z/a).$$

Properties of the z -transform (2)

Theorem. Let x, y, h denote signals $\mathbb{Z} \rightarrow \mathbb{C}$ with associated z -transforms X, Y, H . Then the following properties hold:

4. *Conjugation-property:* let $y(n) = \overline{x(n)}$, then $D_y = D_x$ and

$$Y(z) = \overline{X(\bar{z})}.$$

5. *Time-reversal:* Let $y(n) = x(-n)$, then $D_y = D_x^{-1}$ and

$$Y(z) = X(z^{-1}).$$

6. *Convolution theorem:* Let $y = h * x$, then $D_h \cap D_x \subseteq D_y$ and

$$Y(z) = H(z) \cdot X(z).$$

Proof: Exercise!

Transfer function

Definition. The z -transform $H(z) = \sum_{n \in \mathbb{Z}} h(n)z^{-n}$ of the impulse response h of a continuous LTI-system T , $T[x] = h * x$, is called the *transfer-function* of the system.

Remarks.

- For $h \in \ell^1(\mathbb{Z})$, the Fourier transform $\hat{h}$ coincides with the z -transform of h restricted to the unit circle:

$$\hat{h}(\omega) = \sum_{n \in \mathbb{Z}} h(n)e^{-2\pi i \omega n} = H(e^{2\pi i \omega}).$$

- By abuse of notation, we sometimes write $H(\omega)$ instead of $H(e^{2\pi i \omega})$.